Gastroesophageal Reflux Disease

92

Overview

- ☑ Gastroesophageal reflux (GER) is the passive regurgitation of gastric contents retrograde into the esophagus. It is a physiological process in infants with peak incidence at 6 months of age (50%) and resolves by 2 years of age (10%).
- ☑ GER disease (GERD) is a pathological process when GER leads to troublesome symptoms or complications **(Table 1)** seen in 22–25% of infants and 5–8% of older children.
- ☑ Refractory GERD—GERD not responding to optimal treatment after 8 weeks.

TABLE 1: Gastroesophageal reflux (GER) vs. GER disease (GERD).

GER	***GERD***
No weight loss	Significant weight loss
No respiratory symptoms	Apnea, cyanosis, wheeze, recurrent aspiration pneumonia, and apparent life-threatening event (ALTE)
No symptoms of esophagitis	Dysphagia and lower chest pain
No neurobehavioral symptoms	Sandifer syndrome (abnormal neck hyperextension with torticollis)

Risk Factors

- ☑ *Infants*:
  - Prematurity
  - Medications
  - Congenital anomalies of respiratory tract
  - Postoperative esophageal atresia/diaphragmatic hernia
  - Neurological impairment
  - Abdominal wall defects—gastroschisis and omphalocele
  - Naso/orogastric tube placement
- ☑ *Children*:
  - Obesity
  - Chronic respiratory disorders/wheezing/asthma
  - Neurological impairment
  - Hiatus hernia
  - Post-achalasia treatment
  - Diet—aerated drinks and spicy and fatty meal habits

Clinical Manifestations

- ☑ *Esophageal*: Chest pain, heartburn, odynophagia, and dysphagia
- ☑ *Extraesophageal*: Apnea/cyanotic episodes, stridor, cough, laryngitis, wheezing, recurrent aspiration pneumonia, and Sandifer syndrome
- ☑ *Complications*: Peptic stricture and Barrett's esophagus

- ☑ GER is physiological in most infants and needs no tests for documentation.
- ☑ In GERD, investigations depend on the clinical manifestations.

History and Clinical Examination

- ☑ Reliability of 60–70% in adolescents and older children.
- ☑ Less reliable in infants and in those with extraesophageal symptoms.

Upper Gastrointestinal Endoscopy

- ☑ When typical symptoms are present, upper gastrointestinal endoscopy (UGIE) must be the first investigation.
- ☑ Direct visualization of esophageal mucosa and to obtain biopsies to rule out conditions such as eosinophilic esophagitis.
- ☑ Endoscopically visible damage to the distal esophageal mucosa is the most reliable sign of reflux esophagitis.
- ☑ Typical features of GERD on esophageal histology are basal zone hyperplasia (>20% of total thickness) and elongation of papillae or rete pegs (>50% of total thickness). These may be useful in nonerosive reflux disease (NERD), but they lack sufficient sensitivity and specificity.
- ☑ Endoscopy can also rule out conditions such as achalasia, hiatus hernia, and complications like stricture or Barrett's metaplasia.
- ☑ *Indications for endoscopy:*
  - Hematemesis
  - Dysphagia
  - Feeding aversion
  - History of regurgitation
  - Persistent faltering growth associated with overt regurgitation
  - Back arching or features of Sandifer syndrome
  - Persistent retrosternal or epigastric pain refractory to trial of proton pump inhibitor (PPI)
  - Unexplained iron deficiency anemia
  - Referral for fundoplication

Diagnosis

24-hour Esophageal pH Monitoring

- ☑ Measures the magnitude of acidic reflux over 24 hours using a pH probe placed at the lower esophagus
- ☑ Helps in establishing a temporal association between the patient's symptom episodes and acid reflux
- ☑ Reflux index (percentage of time esophageal pH < 4) of >10% in infants and >7% in children is considered significant
- ☑ Currently not considered ideal for routine use after the advent of impedance pH-metry

Combined Esophageal Multichannel Intraluminal Impedance and pH Monitoring

- ☑ Multichannel intraluminal impedance (MII) measures electrical impedance changes as a bolus of solid, liquid, or air passes between the sensors along the catheter (air > liquid) by incorporating a pH electrode into the impedance catheter.
- ☑ Monitors acidic and nonacidic reflux (pH > 7) which is important, since 50% of children with GERD will be missed, if only acidic reflux is measured.
- ☑ MII-pH also provides details about the duration and degree of the reflux as well as the contents of the refluxate, whether liquid, gas, or both together.
- ☑ It may also be of help to correlate extraesophageal symptoms with reflux with extraesophageal symptoms (nonepileptic seizure-like events or apneas).

Scintigraphy

- ☑ Scintigraphy or milk-scan is a radionuclide-based study for the diagnosis of pulmonary aspiration in GERD
- ☑ Has low sensitivity, lacks standardized technique, and has no accepted normal values
- ☑ May be done to confirm pulmonary aspiration in patients with refractory respiratory symptoms
- ☑ A normal scintigraphy does not always rule out aspiration

Diagnosis

Upper Gastrointestinal Contrast Study

- ☑ Helps in evaluation of anatomic abnormalities which can mimic GERD—hiatal hernia, duodenal web, achalasia, or malrotation of the gut
- ☑ Videofluoroscopic swallow studies differentiate oropharyngeal dysphagia from GERD, especially in neurologically impaired children with recurrent aspiration
- ☑ Not useful for confirming GERD in infants or children

Management

Feed thickeners:

- ☑ Commercial infant feed thickeners are currently not available in India
- ☑ Can use rice cereal as a cheap and acceptable alternative

Feed volume and position:

- ☑ Avoid overfeeding
- ☑ Supine position is better and safe in infants due to risk of sudden infant death syndrome in prone or left lateral position
- ☑ Head-end elevation is not advisable in infants due to increase in intra-abdominal pressure
- ☑ In older children and adolescents, head-end elevation and left lateral position are advised

Lifestyle modification:

- ☑ Avoid chocolates, passive smoking, high fat meals, and late dinner
- ☑ Discourage rapid weight gain

Hypoallergenic formula:

- ☑ 2–4 weeks trial of hypoallergenic formula (extensively hydrolyzed cow milk formula or amino acid-based formula) is recommended in babies who are not breastfed if there is a strong suspicion of milk allergy.
- ☑ To continue breastfeeding with mother being on strict dairy-free diet as a trial in infants who are breastfed.

Management

Pharmacological treatment:

- ☑ *PPIs (lansoprazole and omeprazole)*: Given 30–60 minutes prior to meals. Trial recommended. Duration 4–8 weeks.
- ☑ *H_2-receptor antagonist (ranitidine)*: As an alternative to PPI. Ease of administration as liquid preparation is available
- ☑ *Prokinetics (domperidone and metoclopramide)*:
  - Efficacy not proved
  - Extrapyramidal side effects are a consideration
- ☑ *$GABA_B$ receptor agonist (baclofen):*
  - Reduces the number of transient lower esophageal sphincter (LES) relaxations and accelerates gastric emptying
  - Not recommended due to side effects—lower seizure threshold, drowsiness, and fatigue
- ☑ *Conventional antacids:*
  - Low efficacy and side effects—aluminum toxicity, milk-alkali syndrome (calcium preparations)—not recommended
- ☑ *Surface agents (alginates and sucralfate):*
  - Alginates form a "foam raft" on the top of gastric contents
  - Available in combination with conventional antacids
  - Dose recommended—10–20 mL/dose (after meals) in children ≥12 years old
  - Sucralfate (sucrose–sulfate–aluminum surface agent)—not indicated in routine management of GERD in children

Refractory Gastroesophageal Reflux Disease

- ☑ GERD unresponsive to 8 weeks of optimal treatment
- ☑ Seen in neurologically impaired or post-esophageal surgery
- ☑ Usually have associated esophageal dysmotility, delayed gastric emptying, and swallowing dysfunction and may require antireflux surgery

Surgical Treatment

Indications:

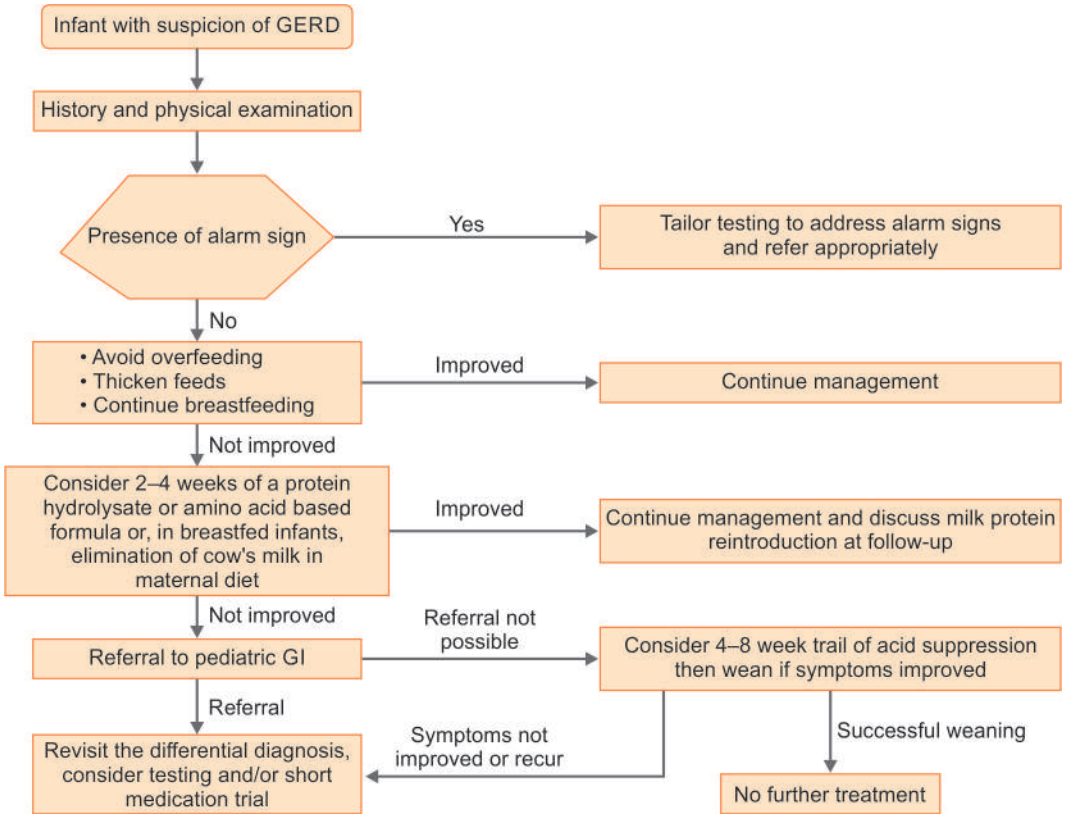
- ☑ Failed medical management
- ☑ GERD-related apparent life-threatening events (ALTEs) in infants
- ☑ Persisting troublesome symptoms despite adequate medical therapy
- ☑ Inability to withdraw medications
- ☑ Complications—recurrent peptic strictures and Barrett's esophagus

Surgery:

- ☑ *Fundoplication:* Laparoscopic or open
- ☑ 86% median success rate for improvement of esophagitis-related symptoms
- ☑ Neurologically impaired children can have more postoperative complications (47%), higher mortality (13%), and failure rate (18%)

Approach Algorithms

Flowchart 1: Management of symptomatic infant.



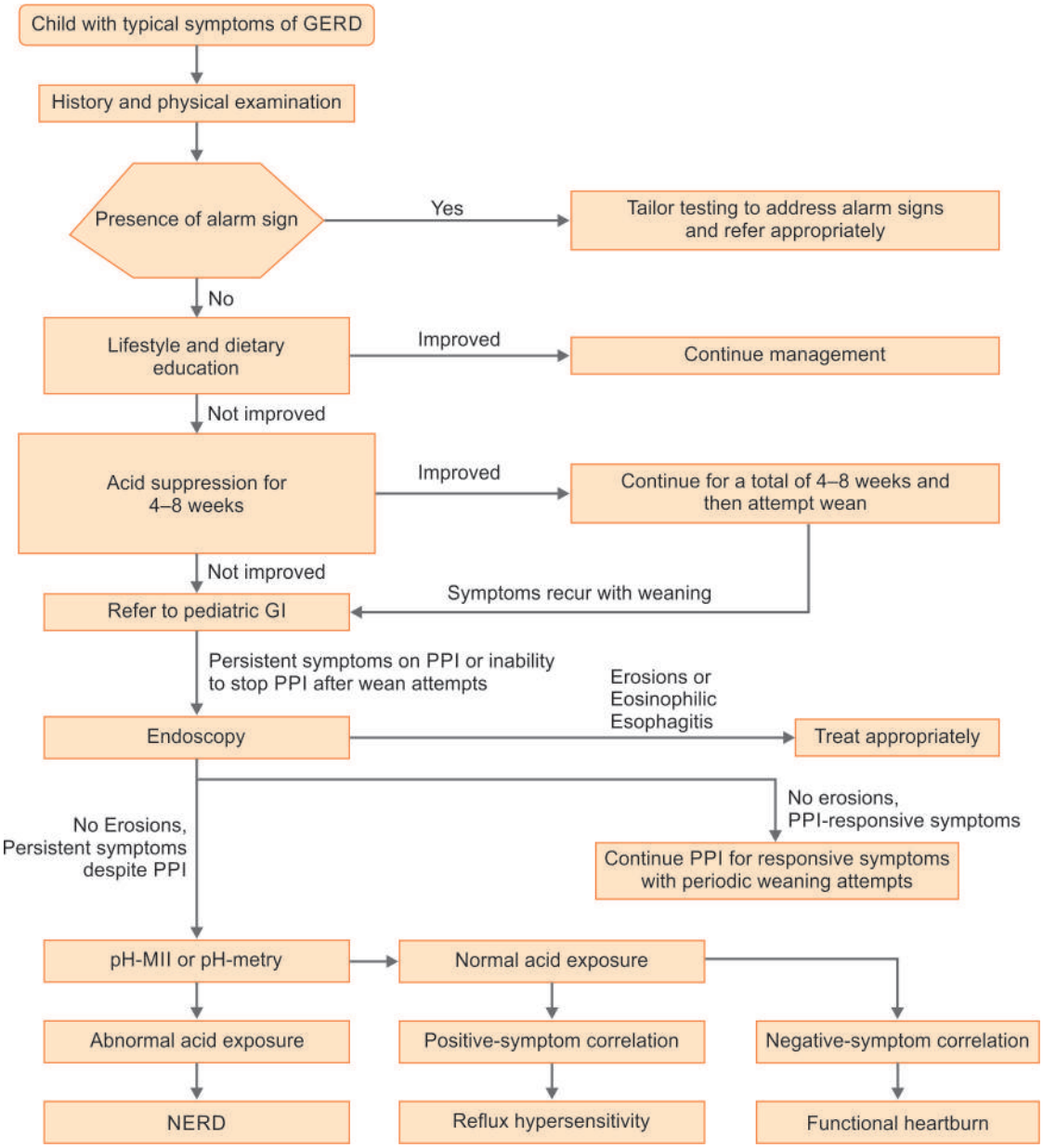
(GERD: gastroesophageal reflux disease; GI: gastrointestinal)

Management

Approach Algorithms

☑ Diagnosis and treatment for typical reflux symptoms in older child

Flowchart 2: Diagnosis and treatment for older child.



(GERD: gastroesophageal reflux disease; GI: gastrointestinal; MII: multichannel intraluminal impedance; NERD: nonerosive reflux disease; PPI: proton-pump inhibitor)

Further Reading

- ☑ Mohan N, Matthai J, Bolia R, Agarwal J, Shrivastava R, Borkar VV, et al. Diagnosis and Management of Gastroesophageal Reflux Disease in Children: Recommendations of Pediatric Gastroenterology Chapter of Indian Academy of Pediatrics, Indian Society of Pediatric Gastroenterology, Hepatology and Nutrition (ISPGHAN). Indian Pediatr. 2021;58:1059-61.
- ☑ Rosen R, Vandenplas Y, Singendonk M, Cabana M, DiLorenzo C, Gottrand F, et al. Pediatric Gastroesophageal Reflux Clinical Practice Guidelines: Joint recommendations of NASPGHAN and ESPGHAN. J Pediatr Gastroenterol Nutr. 2018;66:516-54.

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STANDARD TREATMENT GUIDELINES 2022

Anticipatory Guidance in Adolescents

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